

Lecture 10, Jan 13, 2025.

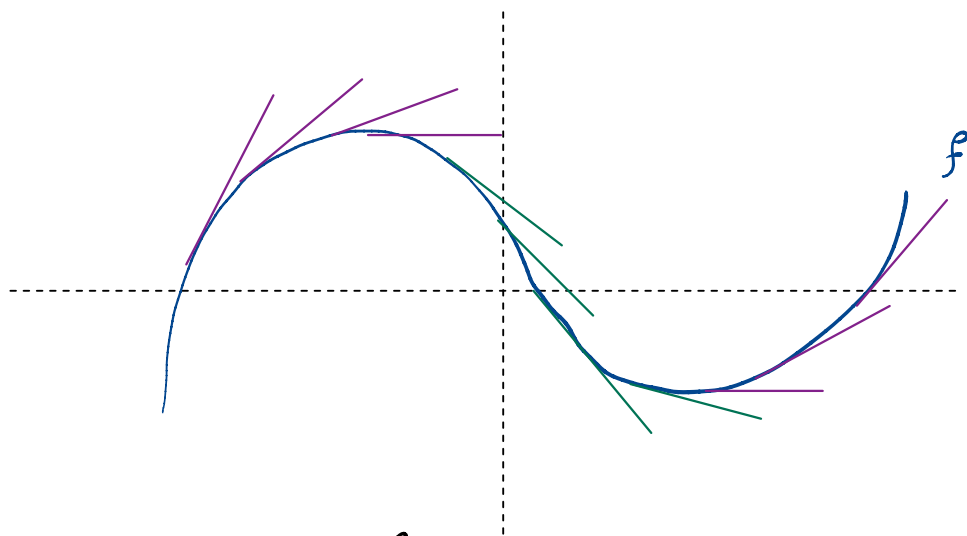
3.3 What Derivatives tell us about the shape of a Graph.

* What Does f' say about f ?

Increasing / Decreasing test.

a). If $f'(x) > 0$ on an interval, then f is increasing on that interval. 

b). If $f'(x) < 0$ on an interval, then f is decreasing on that interval. 



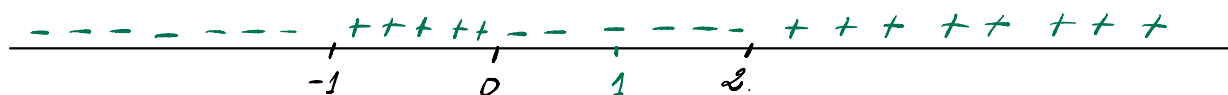
Example. Find where the function $f(x) = 3x^4 - 4x^3 - 12x^2 + 5$ is increasing and where is decreasing.

Sln. $f'(x) = 12x^3 - 12x^2 - 24x$

$$f'(x) > 0 \Leftrightarrow 12x^3 - 12x^2 - 24x > 0$$

$$12x(x^2 - x - 2) > 0$$

$$12x(x - 2)(x + 1) > 0$$



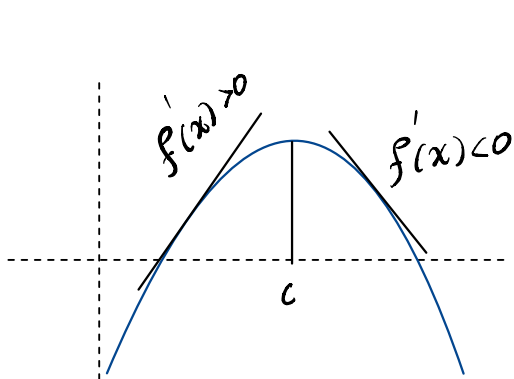
$f(x)$ is increasing on $(-1, 0) \cup (2, \infty)$.

$f(x)$ is decreasing on $(-\infty, -1) \cup (0, 2)$.

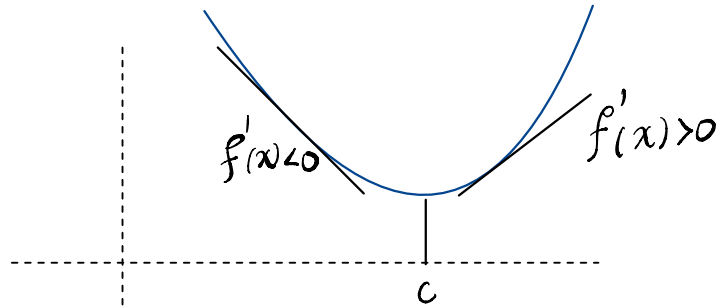
The First Derivative test.

The First Derivative test: Suppose that c is a critical number of a continuous function f .

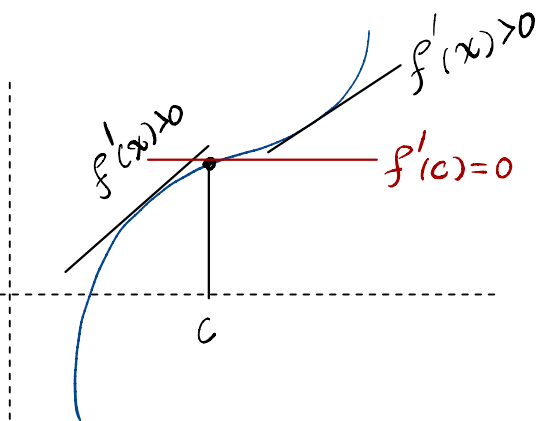
- If f' changes from $(+)$ to $(-)$ at c , then f has a local maximum.
- If f' changes from $(-)$ to $(+)$ at c , then f has a local minimum.
- If f' is positive to the left and right of c , or negative to the left and right of c , then f has no local minimum or maximum.



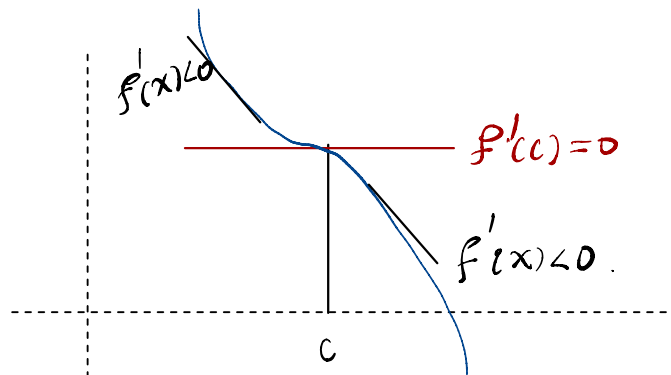
local maximum at c .



local minimum at c .



No maximum or minimum



No maximum or minimum

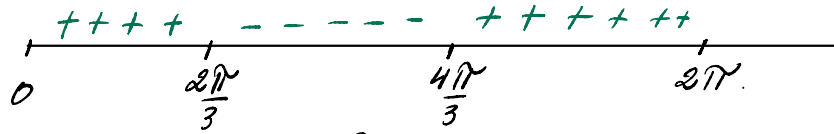
Example. Find the local maximum and minimum values of the function

$$g(x) = x + 2\sin x, \quad 0 \leq x \leq 2\pi.$$

Sln. We need the critical values: then, the derivative is.

$$g'(x) = 1 + 2\cos x.$$

So, $g'(x) = 0 \Leftrightarrow \cos x = -\frac{1}{2}$. the solution of this equation are $\frac{2\pi}{3}, \frac{4\pi}{3}$ then, computing the maximum and minimum.



Because $g'(x)$ change from (+) to (-) at $\frac{2\pi}{3}$, the First Derivative test tell us there is a local maximum at $\frac{2\pi}{3}$ and the local maximum is

$$g\left(\frac{2\pi}{3}\right) = \frac{2\pi}{3} + 2\sin\left(\frac{2\pi}{3}\right) = \frac{2\pi}{3} + 2\left(\frac{\sqrt{3}}{2}\right) = \frac{2\pi}{3} + \sqrt{3} \approx 3.83.$$

Likewise, $g'(x)$ changes from (-) to (+) at $\frac{4\pi}{3}$, then

$$g\left(\frac{4\pi}{3}\right) = \frac{4\pi}{3} + 2\sin\left(\frac{4\pi}{3}\right) = \frac{4\pi}{3} - \sqrt{3} \approx 2.46.$$

is a local minimum.

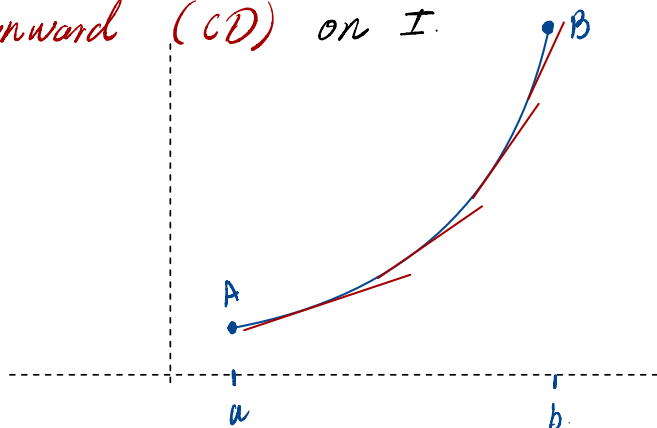
$$g(0) = 0.$$

$$g(2\pi) = 2\pi + 2\sin(2\pi) = 2\pi.$$

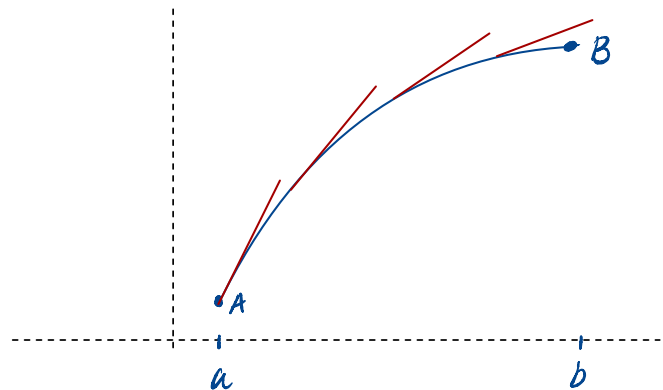
So, the absolute maximum is 2π and absolute minimum is 0.

* What Does f'' say about f ?

Def. If the graph of f lies above all of its tangents on an interval I , then f is called **concave upward (CU)** on I and if the graph of f lies below of its tangents on I , then f is called **concave downward (CD)** on I .



a). Concave upward (CU)



b). Concave downward (CD).

Concavity test.

a). If $f''(x) > 0$ on an interval I , then the graph of f is (CU) on I .

b). If $f''(x) < 0$ on an interval I , then the graph of f is (CD) on I .

Def. A point P on a curve $y=f(x)$ is called an inflection point if f is continuous there and the curve changes from (CU) to (CD) or from (CD) to (CU).

Remk. $f''(x)$ changes from (+) to (-) or (-) to (+).

Example sketch a possible graph of a function f that satisfies the following conditions.

a). $f(0)=0$, $f(2)=3$, $f(4)=6$, $f'(0)=f'(4)=0$.

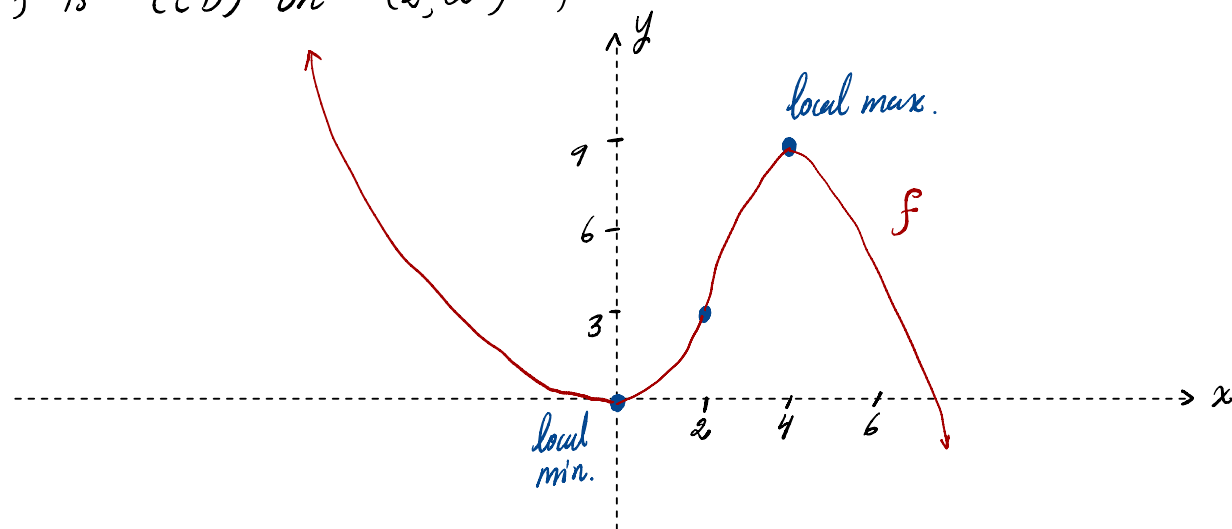
b). $f'(x) > 0$ for $0 < x < 4$, $f'(x) < 0$ for $x < 0$ and $x > 4$.

c). $f''(x) > 0$ for $x < 2$, $f''(x) < 0$ for $x > 2$.

Sln. f is increasing on $(0,4)$ } $x=0$ we have a local min.

f is decreasing on $(-\infty, 0) \cup (4, \infty)$ } $x=4$ we have a local max.

f is (CU) on $(-\infty, 2)$ } $x=2$ is an inflection point.



The Second Derivative test. Suppose f'' is continuous near c .

- a). If $f'(c)=0$ and $f''(c)>0$ then f has a local minimum at c .
 b). If $f'(c)=0$ and $f''(c)<0$ then f has a local maximum at c .

Rmk. The second derivative test is inconclusive when $f''(c)=0$. In other words, at such a point there might be a maximum, there might be a minimum or there might be neither.

Example (for you). Discuss the curve $y=x^4-4x^3$ with respect to concavity points of inflection, and local min/max.

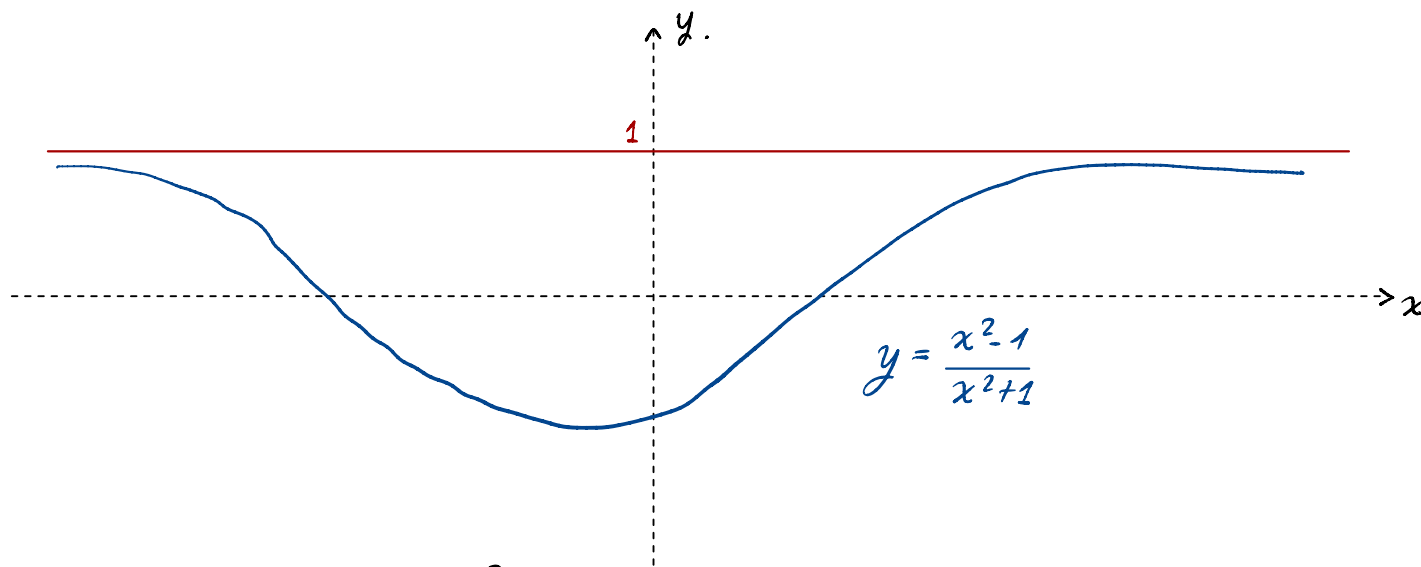
3.4. Limits at infinity; Horizontal Asymptotes.

Let consider

$$f(x) = \frac{x^2-1}{x^2+1}$$

Using the next Data.

x	0	± 1	± 3	± 10	± 100	--- $x \rightarrow \infty$
$f(x)$	-1	0	0.8	0.98019	0.999800	--- $f(x) \rightarrow 1$



then, we have $\lim_{x \rightarrow \infty} f(x) = 1$ and $\lim_{x \rightarrow -\infty} f(x) = 1$.

Def. Let f be a function defined on an interval (a, ∞) . Then,

$$\lim_{x \rightarrow \infty} f(x) = L.$$

means that the values of $f(x)$ can be made arbitrarily close to L by taking x to be sufficiently large.

Def. the line $y = L$ is called a horizontal asymptote of the curve $y = f(x)$ if either $\lim_{x \rightarrow \infty} f(x) = L$ or $\lim_{x \rightarrow -\infty} f(x) = L$.

Thm. If $r > 0$ is rational number then

$$\lim_{x \rightarrow \infty} \frac{1}{x^r} = 0.$$

If $r > 0$ is a rational number s.t x^r is defined for all x , then

$$\lim_{x \rightarrow -\infty} \frac{1}{x^r} = 0.$$

Example. Evaluate the following limit.

$$\lim_{x \rightarrow \infty} \frac{3x^2 - x - 2}{5x^2 + 4x + 1}.$$

Sln.

$$\lim_{x \rightarrow \infty} \frac{3x^2 - x - 2}{5x^2 + 4x + 1} = \lim_{x \rightarrow \infty} \frac{\frac{3x^2 - x - 2}{x^2}}{\frac{5x^2 + 4x + 1}{x^2}}.$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{3x^2}{x^2} - \frac{x}{x^2} - \frac{2}{x^2}}{\frac{5x^2}{x^2} + \frac{4x}{x^2} + \frac{1}{x^2}}.$$

$$= \lim_{x \rightarrow \infty} \frac{3 - \frac{1}{x} - \frac{2}{x^2}}{5 + \frac{4}{x} + \frac{1}{x^2}}.$$

By the thm.

$$= \frac{3}{5}.$$

then, $y = \frac{3}{5}$ is an horizontal asymptote.

If degree Numerator = degree Denominator $\Rightarrow$ the limit is a number.

If degree Numerator > degree Denominator $\Rightarrow$ the limit DNE ∞ or $-\infty$.

If degree Numerator < degree Denominator $\Rightarrow$ the limit exist and is equal to 0.

Example. Find the horizontal asymptotes of the graph of the function

$$f(x) = \frac{\sqrt{2x^2+1}}{3x-5}$$

Sln. • Identify the large power in the Denominator or Numerator.
• Divides the numerator and denominator by this power.

The large power is x .

$$\begin{aligned}\lim_{x \rightarrow \infty} \frac{\sqrt{2x^2+1}}{3x-5} &= \lim_{x \rightarrow \infty} \frac{\frac{\sqrt{2x^2+1}}{x}}{\frac{3x-5}{x}} \\ &= \lim_{x \rightarrow \infty} \frac{\sqrt{\frac{2x^2+1}{x^2}}}{\frac{3x}{x} - \frac{5}{x}} \\ &= \lim_{x \rightarrow \infty} \frac{\sqrt{2 + 1/x^2}}{3 - 5/x} = \frac{\sqrt{2}}{3}.\end{aligned}$$

So, $y = \frac{\sqrt{2}}{3}$ is a horizontal Asymptote of the graph of f .

$$\begin{aligned}\lim_{x \rightarrow -\infty} \frac{\sqrt{2x^2+1}}{3x-5} &= \lim_{x \rightarrow -\infty} \frac{\frac{\sqrt{2x^2+1}}{-x}}{\frac{3x-5}{-x}} \quad \text{b/c } |x| = -x \quad \sqrt{x^2} = |x| \\ &= \lim_{x \rightarrow -\infty} \frac{-\sqrt{\frac{2x^2+1}{x^2}}}{\frac{3x}{-x} - \frac{5}{-x}} = \lim_{x \rightarrow -\infty} \frac{-\sqrt{2 + 1/x^2}}{3 - 5/x} = -\frac{\sqrt{2}}{3}\end{aligned}$$

So, $y = -\frac{\sqrt{2}}{3}$ is a horizontal asymptote.

Example. Compute $\lim_{x \rightarrow \infty} (\sqrt{x^2+1} - x)$.

Sln.

$$\lim_{x \rightarrow \infty} (\sqrt{x^2+1} - x) \times \frac{(\sqrt{x^2+1} + x)}{(\sqrt{x^2+1} + x)} = \lim_{x \rightarrow \infty} \frac{x^2+1 - x^2}{\sqrt{x^2+1} + x} = \lim_{x \rightarrow \infty} \frac{1}{\sqrt{x^2+1} + x} = 0$$

Example. Compute $\lim_{x \rightarrow \infty} \sin\left(\frac{1}{x}\right)$.

Sln. Since $\frac{1}{x} \rightarrow 0$ as $x \rightarrow \infty$ then $\sin\left(\frac{1}{x}\right) \rightarrow \sin(0)$ as $x \rightarrow \infty$.

$$\lim_{x \rightarrow \infty} \sin\left(\frac{1}{x}\right) = \sin(0) = 0.$$

Infinite Limits at infinity.

The notation $\lim_{x \rightarrow \infty} f(x) = \infty$ is used to indicate that the values of $f(x)$ becomes large as x becomes large. Similarly for

$$\lim_{x \rightarrow -\infty} f(x) = \infty, \quad \lim_{x \rightarrow \infty} f(x) = -\infty, \quad \lim_{x \rightarrow -\infty} f(x) = -\infty.$$

Example. Find $\lim_{x \rightarrow \infty} x^3$ and $\lim_{x \rightarrow -\infty} x^3$.

Sln. $\lim_{x \rightarrow \infty} x^3 = \infty$ $\lim_{x \rightarrow -\infty} x^3 = -\infty$.

Example. $\lim_{x \rightarrow \infty} (x^2 - x)$.

Sln. $\lim_{x \rightarrow \infty} (x^2 - x) = \lim_{x \rightarrow \infty} x(x-1) = \infty$.

Example. $\lim_{x \rightarrow \infty} \frac{x^2+x}{3-x} = \lim_{x \rightarrow \infty} \frac{x(x+1)}{x\left(\frac{3}{x}-1\right)} = \lim_{x \rightarrow \infty} \frac{x+1}{\frac{3}{x}-1} = -\infty$.

Example. Sketch the graph of $y = (x-2)^4(x+1)^3(x-1)$ by finding its intercepts and its limits as $x \rightarrow \infty$ and as $x \rightarrow -\infty$.

Sln. For y-intercepts: $f(0) = (-2)^4(1)^3(-1) = -16$; $(0, -16)$.

x-intercepts: $0 = (x-2)^4(x+1)(x-1) \Rightarrow x=2, x=\pm 1$.

So, $(2,0), (1,0), (-1,0)$ are the x-intercepts.

$$\lim_{x \rightarrow \infty} (x-2)^4(x+1)(x-1) = \infty$$

$$\lim_{x \rightarrow -\infty} (x-2)^4(x+1)(x-1) = \infty$$

